

Forecast Comparison VAR vs RF vs BART

```
R_packages <- c("readr", "dplyr", "tidyr", "tibble",
               "ggplot2", "lubridate", "knitr", "zoo")

options(repos = c(CRAN = "https://cloud.r-project.org"))

if (!requireNamespace("librarian", quietly = TRUE)) {
  install.packages("librarian")
}

librarian::shelf(R_packages)
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
if (!requireNamespace("tinytex", quietly = TRUE)) install.packages("tinytex")
if (Sys.which("pdflatex")=="") {library(tinytex); tinytex::install_tinytex()}
theme_set(theme_bw())
knitr::opts_knit$set(root.dir = getwd())
options(knitr.kable.NA = "", knitr.kable.auto_format = FALSE)
knitr::opts_knit$set(kable.auto_number = FALSE)
```

```
### Read csv for previous notebook: VAR_final_with_aplication, RF_application, etc
var_short <- read_csv("VAR_large_all_variables_short.csv", show_col_types = FALSE) |>
  mutate(model = "VAR", horizon_group = "short")

var_long <- read_csv("VAR_large_all_variables_long.csv", show_col_types = FALSE) |>
  mutate(model = "VAR", horizon_group = "long")

rf_short <- read_csv("RF_all_variables_short.csv", show_col_types = FALSE) |>
  mutate(model = "Random Forest", horizon_group = "short")

rf_long <- read_csv("RF_all_variables_long.csv", show_col_types = FALSE) |>
  mutate(model = "Random Forest", horizon_group = "long")

bart_short <- read_csv("BART_all_variables_short.csv", show_col_types = FALSE) |>
  mutate(model = "BART", horizon_group = "short")

bart_long <- read_csv("BART_all_variables_long.csv", show_col_types = FALSE) |>
  mutate(model = "BART", horizon_group = "long")
```

```
get_fc <- function(df, var_name, horizons) {
  df %>%
    filter(variable == var_name, h %in% horizons) %>% # use h column
```

```

    arrange(h) %>%
    pull(forecast)
}

h_short <- 1:3
h_long  <- 1:12

tbl_GDP_short <- tibble(
  h           = h_short,
  `VAR` = get_fc(var_short, "GDP_per_w", h_short),
  `Random Forest` = get_fc(rf_short, "GDP_per_w", h_short),
  BART       = get_fc(bart_short, "GDP_per_w", h_short)
)

tbl_GDP_long <- tibble(
  h           = h_long,
  `VAR` = get_fc(var_long, "GDP_per_w", h_long),
  `Random Forest` = get_fc(rf_long, "GDP_per_w", h_long),
  BART       = get_fc(bart_long, "GDP_per_w", h_long)
)

tbl_CPI_short <- tibble(
  h           = h_short,
  `VAR` = get_fc(var_short, "CPI_ALL_CAN", h_short),
  `Random Forest` = get_fc(rf_short, "CPI_ALL_CAN", h_short),
  BART       = get_fc(bart_short, "CPI_ALL_CAN", h_short)
)

tbl_CPI_long <- tibble(
  h           = h_long,
  `VAR` = get_fc(var_long, "CPI_ALL_CAN", h_long),
  `Random Forest` = get_fc(rf_long, "CPI_ALL_CAN", h_long),
  BART       = get_fc(bart_long, "CPI_ALL_CAN", h_long)
)

tbl_HOUSE_short <- tibble(
  h           = h_short,
  `VAR` = get_fc(var_short, "NHOUSE_P_CAN", h_short),
  `Random Forest` = get_fc(rf_short, "NHOUSE_P_CAN", h_short),
  BART       = get_fc(bart_short, "NHOUSE_P_CAN", h_short)
)

tbl_HOUSE_long <- tibble(
  h           = h_long,
  `VAR` = get_fc(var_long, "NHOUSE_P_CAN", h_long),
  `Random Forest` = get_fc(rf_long, "NHOUSE_P_CAN", h_long),
  BART       = get_fc(bart_long, "NHOUSE_P_CAN", h_long)
)

scale_to_pct <- function(tbl) {
  tbl %>%
    dplyr::mutate(dplyr::across(-h, ~ . * 100))
}

```

```
tbl_GDP_short <- scale_to_pct(tbl_GDP_short)
tbl_GDP_long  <- scale_to_pct(tbl_GDP_long)
tbl_CPI_short <- scale_to_pct(tbl_CPI_short)
tbl_CPI_long  <- scale_to_pct(tbl_CPI_long)
tbl_HOUSE_short <- scale_to_pct(tbl_HOUSE_short)
tbl_HOUSE_long  <- scale_to_pct(tbl_HOUSE_long)

knitr::kable(
  tbl_GDP_short,
  digits = 4,
  caption = "Short-horizon forecasts for GDP per worker (h = 1-3, pct.)"
)
```

Table 1: Short-horizon forecasts for GDP per worker (h = 1-3, pct.)

h	VAR	Random Forest	BART
1	0.0903	0.2803	0.0774
2	0.0730	0.2676	0.0652
3	0.0710	0.2890	0.1528

```
knitr::kable(
  tbl_GDP_long,
  digits = 4,
  caption = "Long-horizon forecasts for GDP per worker (h = 1-12, pct.)"
)
```

Table 2: Long-horizon forecasts for GDP per worker (h = 1-12, pct.)

h	VAR	Random Forest	BART
1	0.0808	0.2533	0.1240
2	0.0554	0.2332	0.1426
3	0.0667	0.2670	0.1560
4	0.0786	0.2877	-0.0286
5	0.0721	0.3001	0.0715
6	0.0716	0.2707	0.1487
7	0.0705	0.2624	0.2114
8	0.0700	0.2595	0.1411
9	0.0695	0.2757	0.0157
10	0.0692	0.2652	0.0190
11	0.0690	0.2786	0.0696
12	0.0688	0.2778	0.1582

```
knitr::kable(
  tbl_CPI_short,
  digits = 4,
  caption = "Short-horizon forecasts for CPI inflation (h = 1-3, pct.)"
)
```

Table 3: Short-horizon forecasts for CPI inflation (h = 1–3, pct.)

h	VAR	Random Forest	BART
1	0.1309	0.2547	0.1711
2	0.1798	0.2734	0.1896
3	0.1748	0.2684	0.1710

```
knitr::kable(
  tbl_CPI_long,
  digits = 4,
  caption = "Long-horizon forecasts for CPI inflation (h = 1-12, pct.)"
)
```

Table 4: Long-horizon forecasts for CPI inflation (h = 1–12, pct.)

h	VAR	Random Forest	BART
1	0.2195	0.2304	0.1243
2	0.2437	0.2565	0.1468
3	0.2244	0.2378	0.2045
4	0.2271	0.2423	0.2176
5	0.2293	0.2316	0.2658
6	0.2310	0.2298	0.2032
7	0.2319	0.2419	0.0769
8	0.2326	0.2545	-0.0578
9	0.2332	0.2505	0.1589
10	0.2336	0.2827	0.2089
11	0.2339	0.3079	0.2840
12	0.2342	0.2545	0.1409

```
knitr::kable(
  tbl_HOUSE_short,
  digits = 4,
  caption = "Short-horizon forecasts for house prices (h = 1-3, pct.)"
)
```

Table 5: Short-horizon forecasts for house prices (h = 1–3, pct.)

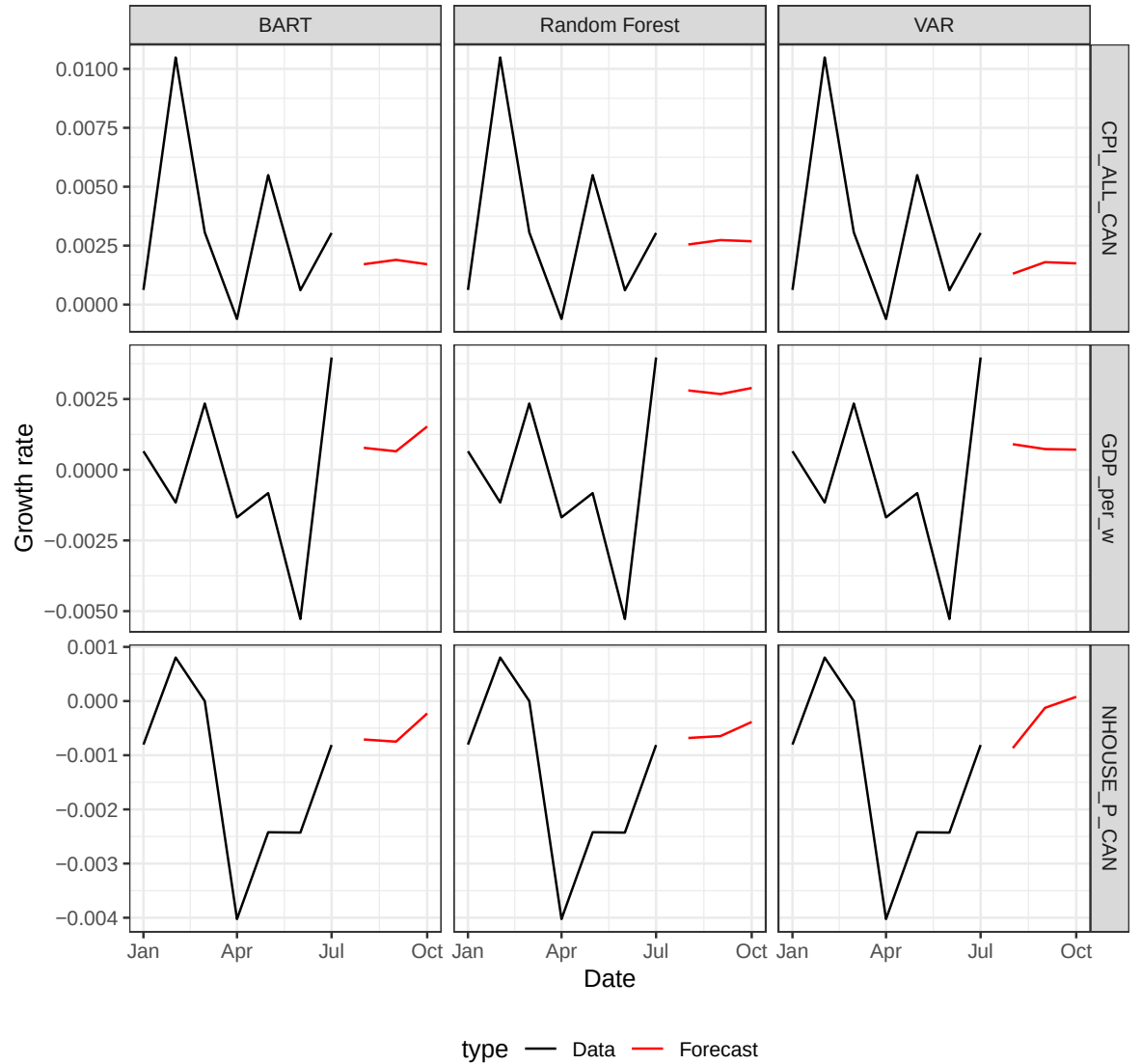
h	VAR	Random Forest	BART
1	-0.0867	-0.0681	-0.0709
2	-0.0125	-0.0646	-0.0749
3	0.0078	-0.0385	-0.0228

```
knitr::kable(
  tbl_HOUSE_long,
  digits = 4,
  caption = "Long-horizon forecasts for house prices (h = 1-12, pct.)"
)
```

Table 6: Long-horizon forecasts for house prices ($h = 1-12$, pct.)

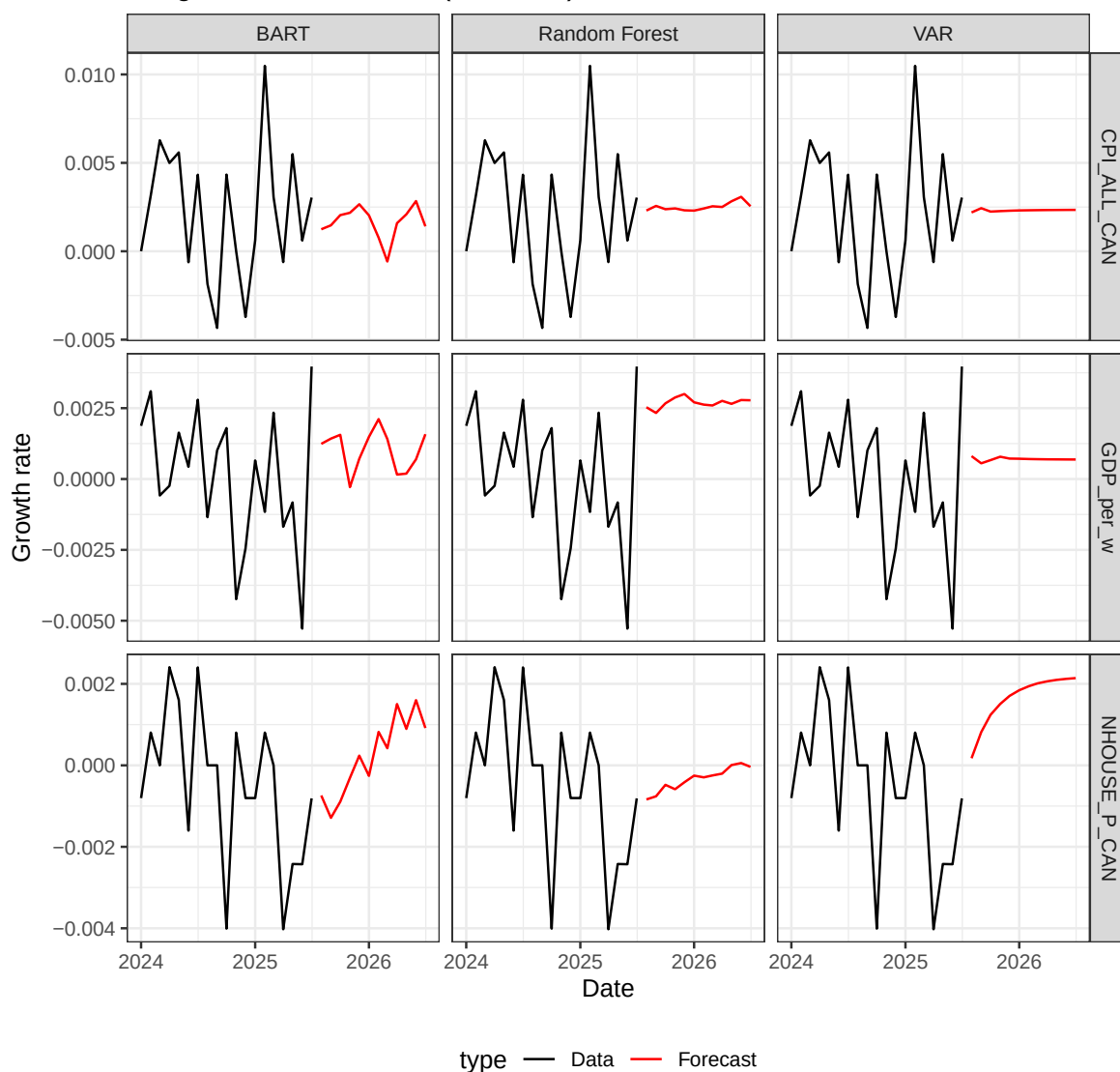
h	VAR	Random Forest	BART
1	0.0174	-0.0837	-0.0740
2	0.0815	-0.0760	-0.1289
3	0.1240	-0.0479	-0.0893
4	0.1507	-0.0587	-0.0307
5	0.1705	-0.0415	0.0238
6	0.1844	-0.0253	-0.0258
7	0.1943	-0.0295	0.0820
8	0.2013	-0.0248	0.0419
9	0.2063	-0.0204	0.1504
10	0.2098	0.0006	0.0894
11	0.2123	0.0055	0.1600
12	0.2141	-0.0038	0.0916

Short-horizon forecasts ($h = 1-3$)



1. Forecast!

Long-horizon forecasts (h = 1–12)



```
comparison <- tribble(
  ~Model,          ~RMSE_1to3,  ~RMSE_1to12,
  "Small VAR (3 vars)", 0.003892, 0.003624,
  "Large VAR (17 vars)", 0.0030008, 0.0034235,
  "Random Forest",    0.002372, 0.002554,
  "BART",             0.00266, 0.002717
)

lag_info <- tribble(
  ~Model,          ~p_short, ~p_long,
  "Small VAR (3 vars)", 7, 7,
  "Large VAR (17 vars)", 2, 1,
  "Random Forest",    12, 5,
  "BART",             11, 3
)
```

```
summary_table <- comparison %>%
  left_join(lag_info, by = "Model") %>%
  select(Model, p_short, RMSE_1to3, p_long, RMSE_1to12)

summary_print <- summary_table
colnames(summary_print) <- c(
  "Model",
  "p (h = 1-3)",
  "RMSE 1-3",
  "p (h = 1-12)",
  "RMSE 1-12"
)

cat("\\setcounter{table}{1}")
```

```
## \setcounter{table}{1}
```

```
kable(
  summary_print,
  digits = 4,
  caption = "Summary of optimal lags and forecast accuracy (RMSE)"
)
```

Table 7: Summary of optimal lags and forecast accuracy (RMSE)

Model	p (h = 1-3)	RMSE 1-3	p (h = 1-12)	RMSE 1-12
Small VAR (3 vars)	7	0.0039	7	0.0036
Large VAR (17 vars)	2	0.0030	1	0.0034
Random Forest	12	0.0024	5	0.0026
BART	11	0.0027	3	0.0027

```
vars_large_17 <- tibble::tribble(
  ~Variable, ~Block, ~Short_Description,
  "GDP_per_w", "Target", "GDP per worker, monthly growth",
  "CPI_ALL_CAN", "Target", "Headline CPI inflation, monthly",
  "NHOUSE_P_CAN", "Target", "National house price index, monthly",
  "IP_new", "Real activity", "Industrial production index, monthly",
  "BSI_new", "Real activity", "Business sentiment / survey index",
  "UNEMP_CAN", "Labour", "Unemployment rate, Canada",
  "EMP_CONS_CAN", "Labour", "Construction employment, Canada",
  "hstart_CAN_new", "Housing", "Housing starts, Canada, monthly",
  "CRED_HOUS", "Credit", "Household / housing credit stock",
  "CRED_HOUS_MORT", "Credit", "Mortgage credit for housing",
  "BANK_RATE_L", "Interest", "BoC policy rate (level)",
  "TBILL_3M", "Interest", "3-month T-bill rate",
  "Exp_BP_new", "External", "Exports of goods (BOP), monthly",
  "Imp_BP_new", "External", "Imports of goods (BOP), monthly",
  "IPPI_METAL_CAN", "Prices", "Metal producer price index, Canada",
  "OILP_new", "Commodities", "Oil price index",
  "WTISPLC", "Commodities", "WTI spot crude oil price (level)"
)
```

```
knitr::kable(
  vars_large_17,
  booktabs = TRUE,
  caption = " The 17 variables selected for the large VAR model"
)
```

Table 8: The 17 variables selected for the large VAR model

Variable	Block	Short_Description
GDP_per_w	Target	GDP per worker, monthly growth
CPI_ALL_CAN	Target	Headline CPI inflation, monthly
NHOUSE_P_CAN	Target	National house price index, monthly
IP_new	Real activity	Industrial production index, monthly
BSI_new	Real activity	Business sentiment / survey index
UNEMP_CAN	Labour	Unemployment rate, Canada
EMP_CONS_CAN	Labour	Construction employment, Canada
hstart_CAN_new	Housing	Housing starts, Canada, monthly
CRED_HOUS	Credit	Household / housing credit stock
CRED_HOUS_MORT	Credit	Mortgage credit for housing
BANK_RATE_L	Interest	BoC policy rate (level)
TBILL_3M	Interest	3-month T-bill rate
Exp_BP_new	External	Exports of goods (BOP), monthly
Imp_BP_new	External	Imports of goods (BOP), monthly
IPPI_METAL_CAN	Prices	Metal producer price index, Canada
OILP_new	Commodities	Oil price index
WTISPLC	Commodities	WTI spot crude oil price (level)

```
write.csv(vars_large_17, "vars_17_large_model.csv", row.names = FALSE)
```